Shri Sant Gajanan Maharaj College of Engineering Shegaon

Electrical Engineering Department

Course Title & Course Code: Electromagnetic Fiels (4EP01)

Class: Second year Electrical Semester: IV

Name of the Course Teacher: DrA.U.Jawadekar

Title of the innovative practice: Think-Pair-Share for Understanding Electric Field Due to a Charged Ring

Objectives/Goals of the practice:

The Think-Pair-Share method is designed to:

- Enhance conceptual understanding of the electric field due to a charged ring.
- Encourage active learning through discussion and peer interaction.
- Develop problem-solving skills by applying Gauss's Law and integrals.

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Use of Appropriate Methods:

Think Phase:

- Students individually analyse a charged ring **system** and predict the field behaviour at different points.

Pair Phase:

- Students discuss their answers with a partner and refine their reasoning.

Share Phase:

- Groups present their observations to the class.
- The discussions is facilitated by showing plots using MATLAB.

Significance of Results:

Encouraged collaborative learning and critical thinking

Effective Presentation:

MATLAB simulation used to plot electric field intensity

The ring is discretized into 100-point charges.

The field is calculated at points in a 3D grid around the ring.

The quiver3 function plots the electric field vectors in 3D.

An animation of the 3D view rotating around the charged ring was created to help better visualize the electric field from different angles

POs Mapped: PO1,PO2,PO3,PO5,PO12

Reflective Critique:

Students appreciated the peer discussions but requested additional examples with different charge distributions.

Faculty Suggested adding numerical problems and a MATLAB-based live demo for better clarity.

Peer Review & Critique Form Link: https://forms.gle/PLPWxUHQpmsi4E4A6

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Peer Review and Critique Form Link:

Evidences of success:

75% of students accurately applied the concept in problem-solving

Challenges faced during implementation:

Some students struggled with mathematical derivations of the electric field

Activity Link :
https://docs.google.com/document/d/1plu3wS9waDnr0gc5tqAMH3Ey62xEoGjJ/edit?usp=sharing&ouid=108857555327832468577&rtpof=true&sd=true

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Example of Think-Pair-Share in Electromagnetic Field Theory (EMF)

Topic: Electric Field Due to a Charged Ring

Step 1: THINK (Individual Reflection)

- Question given to students
  what will be the electric field at the center of a uniformly charged ring? Will it be zero or nonzero? Why?
- Students were given instructions to think individually about the concept using Coulomb's Law and Symmetry Principles**.**
- They were asked to write down their initial answers and reasoning.

Step 2: PAIR (Discuss with a Partner)

- Students paired up and compared their answers and discussed about the problem

Discussion Points:
If their reasoning is correct.
If they have different answers, what assumptions or mistakes might be present?
Can they use **Gauss's Law** to verify their answer?

Step 3: SHARE (Classroom Discussion)

- Each pair shared their conclusion with the class
- Discussion was encouraged and then final ans was explained to students
- Final Answer: *The electric field at the centre of a uniformly charged ring is zero due to symmetry.*
- The discussion was further extended by asking questions

  What happens if we move along the axis of the ring?

  Can you derive the electric field at a point along the axis?

Benefits:

Encourages active learning and critical thinking
Helps student's correct misconceptions through peer discussion
Reinforces problem-solving skills in a collaborative way